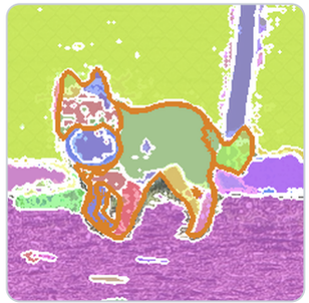
(a) Ask, don't draw: masks become tokens, the MLLM picks one

Query *"Locate the body of the dog."*



SAM 3 → frozen



259 candidate masks

point grid · SAM 3 text

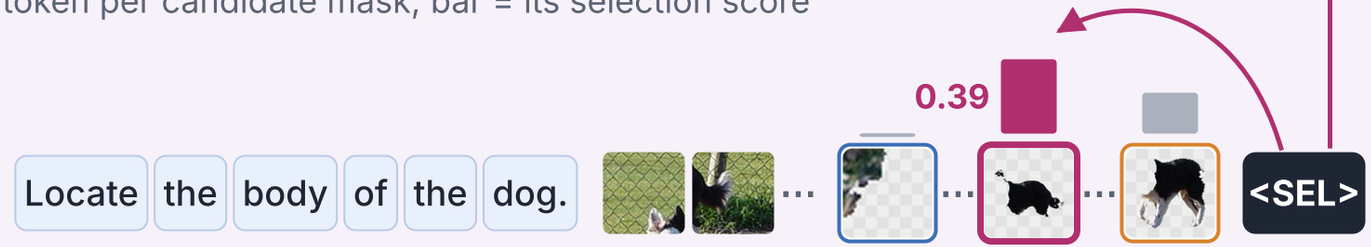
MLLM picks → no mask decoder



selected mask

MLLM input sequence Qwen3-VL-4B, 64.2M trainable parameters

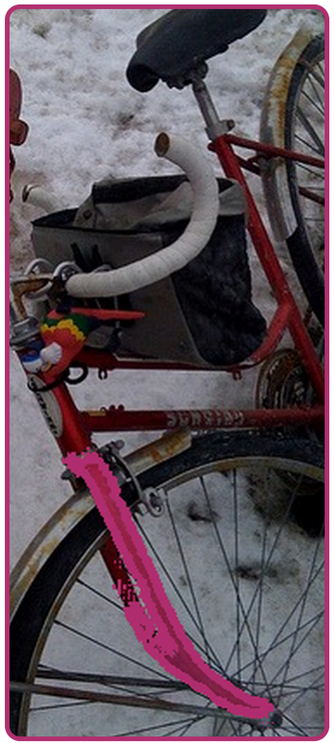
one token per candidate mask; bar = its selection score



(b) One read-out, three kinds of answer

Part

"Locate the fork of the bicycle."



1 mask, IoU 0.68

PACO +14.5

Multi-target

"right man and the cake"



2 masks, IoU 0.95

gRefCOCO +8.0

No target

"man on left"



max σ 0.86 < 0.90

N-acc 94.0%